

White noise analysis of neural coding:

We want to model a visual neuron with an LNP architecture,

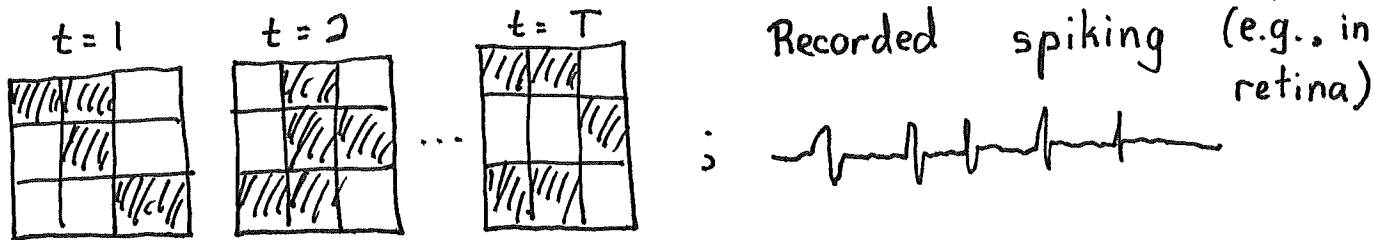
$$k \sim \text{Pois}(\mathcal{F}(L(s)))$$

$$L(s) = \int dx' \int dy' \int dt' K(x_0 - x', y_0 - y', t - t') s(x', y', t')$$

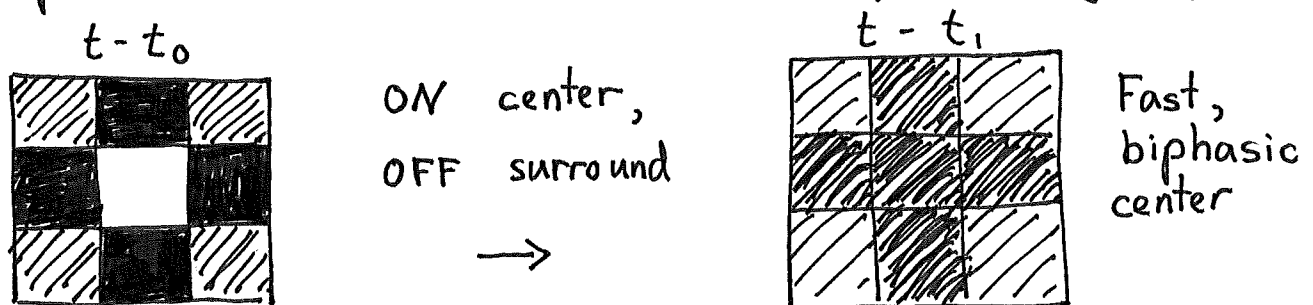
How can we determine K and $\mathcal{F}$? A common approach is to compute the "spike triggered average," or STA.

Standard procedure:

1. Show random "white-noise stimuli" and record spikes.

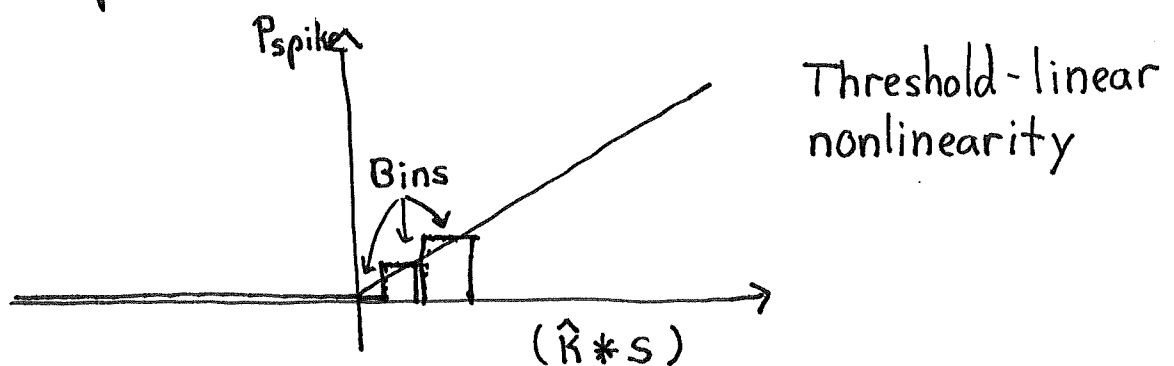


2. Compute the average stimulus preceding spikes



3. Compute convolution of STA and stimulus.

4. Evaluate probability of spiking.



Alternate ways to write $L(s)$:

Continuous space and time:

$$L(s) = \int dx' \int dy' \int dt' \underbrace{K(x', y', t - t')}_{\text{Redefinition that shifts and Flips spatial coordinates}} s(x', y', t')$$

Redefinition that shifts and
Flips spatial coordinates

$$= \int dx' \int dy' \int dt' K(x', y', t') s(x', y', t - t')$$

Pixelated space and time:

$$L(s) = \sum_p K_p S_{pt} \quad ; \quad \begin{array}{l} p: \text{pixel index for kernel} \\ x_p: \text{x-coordinate of pixel } p \\ y_p: \text{y-coordinate} \\ t_p: \text{t-coordinate (looks back in time)} \end{array}$$
$$= K S$$

The pixelated matrix form is tidy and simplifies the derivation. In this notation, the firing rate is

$$r_t = \mathcal{F}(\sum_p K_p S_{pt})$$

Linear neuron:

Suppose $\mathcal{S}(x) = x$, so

$$h_t = \sum_p K_p S_{pt}$$

and the experimenter only needs to estimate K . "Linear system identification."

Choose K to minimize mean squared error

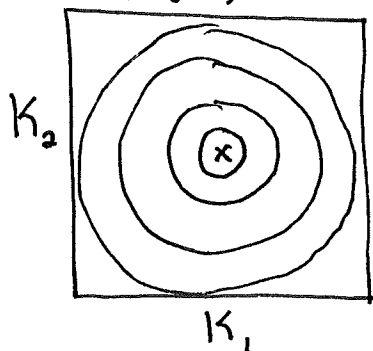
$$\mathcal{L}(K) = \frac{1}{T} \sum_{t=1}^T (h_t - \sum_p K_p S_{pt})^2$$

In machine learning lectures, we showed that the solution to this linear regression problem is

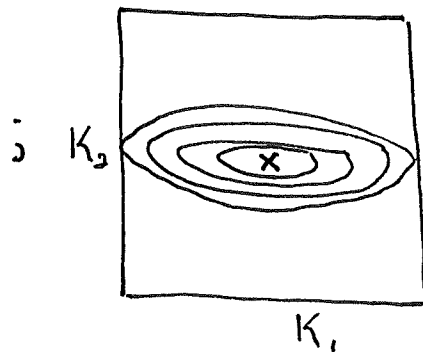
$$\underbrace{\hat{K}}_{p \times 1} = \underbrace{(S S^T)^{-1}}_{p \times p} \underbrace{S h}_{p \times 1}$$

We also explained that error contours are ellipses. To estimate each K_p equally well, want these ellipses to be circles

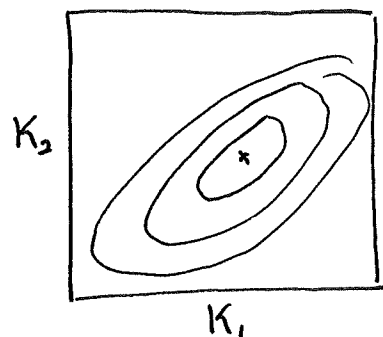
"Good"
 $\mathcal{L}(K)$



K_1 is sloppy



Estimates are correlated



Equivalently, we want second derivative matrix to be isotropic, because this means that $\mathcal{L}(K)$ has equal curvature in all directions.

This matrix is

$$\begin{aligned}\frac{\partial^2 \mathcal{L}}{\partial K_{p_1} \partial K_{p_2}} &= \frac{\partial^2}{\partial K_{p_1} \partial K_{p_2}} \left(\frac{1}{T} \sum_{t=1}^T \left(h_t^2 - 2 \sum_p K_p S_{p,t} h_t \right. \right. \\ &\quad \left. \left. + \sum_p \sum_{p'} K_p S_{p,t} K_{p'} S_{p',t} \right) \right) \\ &= \frac{2}{T} \sum_{t=1}^T S_{p_1,t} S_{p_2,t} \\ &= \frac{2}{T} (S S^T)_{p_1, p_2}\end{aligned}$$

It's isotropic if

$$(S S^T)_{p_1, p_2} = A \delta_{p_1, p_2}, \quad A \text{ constant}$$

Stimuli with this property have completely uncorrelated pixels. It's termed "white noise", and typically taken to be Gaussian

$$S_{x,y,t} \sim \mathcal{N}(0, \sigma_s^2)$$

Thus

$$\begin{aligned}\langle (S S^T)_{p_1, p_2} \rangle &= \left\langle \sum_{t=1}^T S_{p_1,t} S_{p_2,t} \right\rangle = \sum_{t=1}^T \langle S_{x_{p_1}, y_{p_1}, t-t_{p_1}} \\ &\quad S_{x_{p_2}, y_{p_2}, t-t_{p_2}} \rangle = \sum_{t=1}^T \sigma_s^2 \delta_{x_{p_1}, x_{p_2}} \delta_{y_{p_1}, y_{p_2}} \delta_{t_{p_1}, t_{p_2}} \\ &= T \sigma_s^2 \delta_{p_1, p_2}\end{aligned}$$

$$\frac{\partial^2 \mathcal{L}}{\partial K_{p_1} \partial K_{p_2}} \approx 2 \sigma_s^2 \delta_{p_1, p_2}$$

$$(S S^T)^{-1}_{p_1, p_2} \approx \frac{1}{T \sigma_s^2} \delta_{p_1, p_2}$$

For this reason, white noise stimuli are often used for system identification.

The Kernel estimate is simply

$$\hat{K} = (SS^T)^{-1} S h = \frac{1}{T \sigma_s^2} S h$$

h_t is the firing rate, but we measure spikes. We write

$$h_t = \frac{k_t}{\Delta t} ; \quad k_t : \# \text{ spikes in bin } t$$

$$\Delta t : \text{bin width}$$

For small Δt , k_t will be zero or one. Therefore,

$$\hat{K}_p = \frac{1}{T \sigma_s^2} \sum_t S_{pt} h_t = \frac{1}{T \sigma_s^2 \Delta t} \sum_{i=1}^{n_s} S_{pt_i} ; \quad \begin{array}{l} n_s : \# \text{ spikes} \\ t_i : \text{spike times} \end{array}$$

$$= \frac{1}{T \sigma_s^2 \Delta t} \sum_{i=1}^{n_s} S_{x_p, y_p, t_i - t_p}$$

$$\hat{K} = \underbrace{\frac{n_s}{T \sigma_s^2 \Delta t}}_{\text{constant}} \underbrace{\sum}_{\text{"spike-triggered average," STA}}$$

Including nonlinearity:

We now return to the case where

$$h_t = \mathcal{F}\left(\sum_p K_p s_{pt}\right)$$

and we want to infer both K and $\mathcal{F}$. For small enough Δt , $\lambda = h \Delta t$ is small, and we can approximate the Poisson distribution as

$$P(k) = \begin{cases} 1 - \lambda, & k=0 \\ \lambda, & k=1 \end{cases}$$

Therefore, the probability that a neuron spikes is

$$P_{\text{spike}} = P_{\text{spike}}(L(s)) = \Delta t \mathcal{F}(L(s))$$

If we knew $L(s)$, and plotted $P_{\text{spike}}(L(s))$, the resulting curve is proportional to $\mathcal{F}$.

Standard procedure:

1. Show Gaussian white-noise stimuli and record spikes.
2. Evaluate $\hat{K} = \mathcal{L}$
3. Evaluate $\hat{L} = \hat{K} s$
4. $\hat{\mathcal{F}}(\hat{L}) = \frac{1}{\Delta t} P_{\text{spike}}(\hat{L})$

You now have a model neuron to predict spiking responses to new stimuli.

$$P(k_t | s) = \text{Pois}(\hat{\mathcal{F}}(\hat{K} s))$$

This procedure is heuristic, often works very well, but can sometimes fail, even catastrophically.